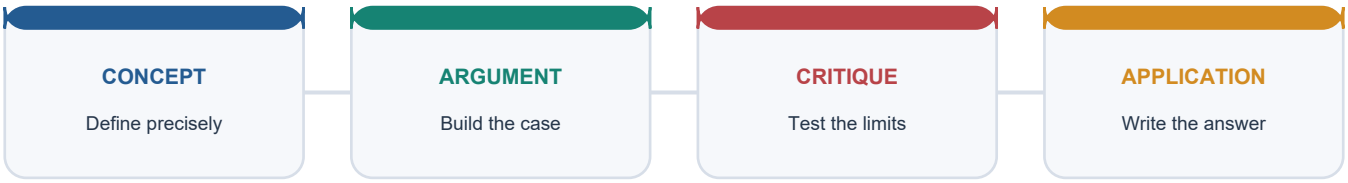
SOLVED PRACTICE WORKBOOK

Tsunami and Coastal Hazard Management - Solved Practice Workbook

GS-III Disaster Management | 32 solved MCQs, PYQ applications and Mains models

LEARNING PATH



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Tsunami and Coastal Hazard Management - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. What is the literal meaning of the Japanese word 'tsunami'?

A. 'Harbour wave'. B. 'Deep trench'. C. 'Rising tide'. D. 'Storm front'.

Answer: A. Explanation:

- A: Correct: VisionIAS glosses tsunami as Japanese for 'harbour wave'.
- B: Incorrect: this invents an unrelated geological-feature gloss.
- C: Incorrect: a tsunami is explicitly distinguished from a tide, so this gloss is wrong.
- D: Incorrect: this borrows storm/cyclone imagery, not the source's actual gloss.

Trap: Do not confuse the etymological gloss with an unrelated oceanographic or meteorological term.

Q2. Which tsunami-impact factor concerns the generating event rather than the receiving coast?

A. The colour-coded alert level issued by the state government. B. Characteristics of the generating seismic event. C. The population density of the district struck. D. The number of fishing harbours on the affected coast.

Answer: B. Explanation:

- A: Incorrect: an alert level is a warning-system output, not a source-side effect-determining factor.
- B: Correct: the generating seismic event's own characteristics is one of the three named effect-determining factors.
- C: Incorrect: population density is an exposure factor, not one of the three named effect-determining factors in this specific list.
- D: Incorrect: harbour count is not named as an effect-determining factor by the source.

Trap: Distinguish the three named effect-determining factors (event characteristics, distance/magnitude, bathymetry) from general exposure or warning-output variables.

Q3. Which tsunami-impact factor concerns the configuration of undersea topography?

A. The literacy rate of the coastal population. B. The number of ports along the affected stretch. C. Bathymetry (ocean depth configuration). D. The month of the year in which the event occurs.

Answer: C. Explanation:

- A: Incorrect: literacy rate is a social-vulnerability variable, not a named effect-determining factor here.
- B: Incorrect: port count is an exposure detail, not a named effect-determining factor.
- C: Correct: bathymetry, meaning ocean depth configuration, is explicitly named as an effect-determining factor.
- D: Incorrect: calendar month is not named as a determining factor for tsunami effect.

Trap: Bathymetry means ocean-depth configuration specifically; do not substitute an unrelated social or infrastructural count.

Q4. Why are submarine-landslide tsunamis plausible in parts of the Bay of Bengal and Arabian Sea?

A. Because both seas are entirely free of any river-sediment input. B. Because both seas lie outside any seismically active zone. C. Because landslide tsunamis are said to be impossible in tropical latitudes. D. Because sediment deposition from the Ganges and Indus rivers makes landslide-triggered tsunamis a plausible regional scenario.

Answer: D. Explanation:

- A: Incorrect: this reverses the source, which specifically cites sediment deposition from these rivers.
- B: Incorrect: the Bay of Bengal/Arabian Sea region is not described as seismically inactive.
- C: Incorrect: the source names landslides as a real trigger mechanism, not an impossible one.

- D: Correct: Ganges/Indus sediment deposition is the named basis for treating landslide-triggered tsunamis as a plausible regional scenario.

Trap: Do not detach the landslide-tsunami risk from its named cause: river-sediment deposition altering seabed stability.

Q5. Which statement accurately presents the document-period description of India's coastline exposure?

A. Both the East and West coastlines are vulnerable, together exceeding 2,200 km of heavily populated shoreline. B. Only the East coastline carries any documented tsunami vulnerability. C. Only the West coastline carries any documented tsunami vulnerability. D. India's coastline is described as sparsely populated end to end.

Answer: A. Explanation:

- A: Correct: the source explicitly names both East and West coastlines as vulnerable, exceeding 2,200 km of heavily populated shoreline -- a document-period historical figure requiring a current INCOIS/GSI source before being treated as today's exact statistic.
- B: Incorrect: this omits the West coast, which the source also names as vulnerable.
- C: Incorrect: this omits the East coast, which the source also names as vulnerable.
- D: Incorrect: the source specifically calls the shoreline heavily populated, not sparsely populated.

Trap: Do not restrict India's named tsunami exposure to only one coastline; the source names both East and West.

Q6. What did the 2004 tsunami demonstrate about population density and damage severity?

A. Damage was heaviest in sparsely populated, elevated coastal stretches. B. Maximum damage in 2004 occurred in low-lying coastal areas with high population density. C. Damage in 2004 was concentrated in high-altitude inland districts far from the coast. D. Population density had no bearing on the severity of damage recorded in 2004.

Answer: B. Explanation:

- A: Incorrect: this reverses the source's low-lying/dense-population damage pattern.
- B: Correct: the source ties maximum 2004 damage to low-lying, densely populated coastal areas.
- C: Incorrect: 2004 damage was coastal, not inland/high-altitude.
- D: Incorrect: the source explicitly links density to damage severity.

Trap: Keep the 2004 damage pattern tied to its two named conditions together: low-lying elevation and high population density.

Q7. Which natural coastal features reduced damage in parts of the 2004 tsunami?

A. Only offshore oil-rig platforms. B. Only large-scale afforestation carried out after 2004. C. Mangroves, forests, sand dunes and coastal cliffs. D. Only artificial breakwaters and concrete sea walls.

Answer: C. Explanation:

- A: Incorrect: oil-rig platforms are not named as a coastal barrier by the source.
- B: Incorrect: this misdates the barrier as post-event afforestation rather than pre-existing natural cover.
- C: Correct: mangroves, forests, sand dunes and coastal cliffs are explicitly named as the best natural barriers in 2004.
- D: Incorrect: the source's 'best barriers' finding is about natural features, not artificial structures.

Trap: Do not substitute artificial structures for the specific natural features (mangroves/forests/dunes/cliffs) the source credits.

Q8. Where was heavy 2004 damage linked to human removal of a natural coastal buffer?

A. Where mangroves were freshly planted immediately before the event. B. Where coral reefs had never been disturbed. C. Where coastal cliffs were reinforced with additional rock. D. Where sand dunes were heavily mined, heavy damage occurred.

Answer: D. Explanation:

- A: Incorrect: fresh mangrove planting is a protective, not a damaging, intervention in the source's account.
- B: Incorrect: undisturbed coral reefs are not linked to heavy damage by the source.
- C: Incorrect: reinforcing cliffs is protective, not the damage-linked interference the source names.
- D: Correct: the source ties heavy damage to areas where sand dunes were heavily mined, removing a natural buffer.

Trap: The damage-linked interference is specifically dune mining, not any generic coastal construction activity.

Q9. Which data combination is central to tsunami vulnerability and risk mapping?

A. Vulnerability/risk mapping using coastal land-use maps and bathymetry data. B. Relying solely on historical casualty counts from the 2004 event. C. Mapping based only on administrative district boundaries. D. Mapping based only on the number of registered fishing vessels.

Answer: A. Explanation:

- A: Correct: NDMA's recommended approach combines coastal land-use maps with bathymetry for vulnerability/risk mapping.
- B: Incorrect: historical casualty counts alone are not the named NDMA risk-mapping method.
- C: Incorrect: administrative boundaries alone are not the named basis for this mapping.
- D: Incorrect: vessel counts are not the named basis for NDMA's risk-mapping recommendation.

Trap: NDMA's method combines land-use maps and bathymetry together; do not substitute an unrelated single dataset.

Q10. Why is seismic detection alone insufficient for a confirmed tsunami assessment?

A. Because earthquakes cannot occur beneath the ocean. B. Because seismic data identify the source, while models and sea-level observations are needed to assess whether a wave was generated and how it may affect coasts. C. Because only rainfall gauges can confirm tsunami generation. D. Because coastal authorities must wait for visible flooding before receiving technical information.

Answer: B. Explanation:

- A: Incorrect: many earthquakes occur beneath the ocean.
- B: Correct: source characterisation, modelling and BPR/tide-gauge observations perform distinct functions in a multi-sensor warning chain.
- C: Incorrect: rainfall gauges do not confirm tsunami waves.
- D: Incorrect: the purpose of early warning is to act before destructive coastal arrival.

Trap: Earthquake detection starts the assessment; it does not by itself confirm a propagating tsunami.

Q11. What function do Bottom Pressure Recorders perform in the warning chain?

A. Licensing coastal construction under the CRZ Notification. B. Administering post-disaster compensation payments. C. Detecting open-ocean wave propagation. D. Issuing the final local evacuation order.

Answer: C. Explanation:

- A: Incorrect: CRZ licensing is a regulatory function unrelated to BPR sensing.
- B: Incorrect: compensation administration is a separate relief function, not a BPR role.
- C: Correct: BPRs detect open-ocean wave propagation as part of the detection chain.
- D: Incorrect: local evacuation orders remain a local-authority function, not a BPR function.

Trap: BPRs are sensing instruments for open-ocean wave detection, not decision-making or regulatory bodies.

Q12. Which engineered measures may be combined with bio-shields for coastal-hazard mitigation?

A. Deep-sea oil exploration platforms. B. Long-distance rail corridors connecting coastal cities. C. Urban metro rail systems in coastal metropolitan areas. D. Submerged sand barriers/dykes and sand dunes with sea-weed planting.

Answer: D. Explanation:

- A: Incorrect: oil platforms are not a named coastal-hazard mitigation measure.
- B: Incorrect: rail corridors are unrelated to the named mitigation measures.
- C: Incorrect: metro systems are unrelated to the named coastal-hazard mitigation measures.
- D: Correct: submerged sand barriers/dykes and sea-weed-planted sand dunes are named structural/ecosystem mitigation measures.

Trap: Only the source's own named structural/ecosystem measures count; do not substitute unrelated infrastructure.

Q13. Which Pondicherry institution is cited as a local-knowledge centre for coastal preparedness?

A. The M.S. Swaminathan Research Foundation. B. The Indian Institute of Technology, Chennai campus. C. The Reserve Bank of India's regional office. D. The Election Commission of India's regional office.

Answer: A. Explanation:

- A: Correct: the M.S. Swaminathan Research Foundation, Pondicherry, is named as a local-knowledge centre example.
- B: Incorrect: this IIT campus is not the named local-knowledge centre in the source.
- C: Incorrect: the RBI regional office is unrelated to coastal local-knowledge centres.
- D: Incorrect: the Election Commission's office is unrelated to coastal disaster-knowledge functions.

Trap: The named local-knowledge centre is the M.S. Swaminathan Research Foundation specifically, not a generic nearby institution.

Q14. How should coral reefs be classified relative to engineered measures such as sea walls?

A. As part of one undifferentiated category, with no distinction from sea walls. B. As an ecosystem-based/natural barrier, distinct from but complementary to engineered structural measures such as sea walls and breakwaters. C. As having no coastal-protection role recognised by the source at all. D. As a purely administrative classification with no physical protective function.

Answer: B. Explanation:

- A: Incorrect: the source's own natural-barrier evidence (mangroves/dunes/reefs performing best in 2004) supports a distinct ecosystem-based category, not an undifferentiated one shared with sea walls.
- B: Correct: coral reefs, alongside mangroves, sand dunes and bio-shields, are ecosystem-based/natural barriers -- complementary to, but distinct from, engineered structural measures such as sea walls, breakwaters and dykes.
- C: Incorrect: the source explicitly credits natural barriers, including reefs, mangroves and dunes, with protective value in the 2004 event.
- D: Incorrect: the classification is protective/physical, not merely administrative.

Trap: Do not lump coral reefs into the structural-measures category; they are an ecosystem-based/natural barrier, per the source's own natural-barrier evidence and Section 8's caution against treating ecosystem-based measures as secondary.

Q15. What preparatory action is needed for structures identified as vulnerable to tsunami impact?

A. A blanket ban on any new coastal construction nationwide. B. Mandatory relocation of all coastal populations inland. C. Identification of vulnerable structures for retrofitting. D. Automatic demolition of every coastal structure over ten years old.

Answer: C. Explanation:

- A: Incorrect: a blanket nationwide ban is not the named CRZ-linked approach, which is category-specific.
- B: Incorrect: mandatory blanket relocation is not the named preparatory measure.
- C: Correct: the source names identification of vulnerable structures for retrofitting as a preparatory step.
- D: Incorrect: automatic demolition by age is not a named measure.

Trap: The named step is identification for retrofitting, not demolition, blanket bans or forced relocation.

Q16. Which groups may support community search and rescue after a tsunami?

A. Only the armed forces, with no role for community-based organisations. B. Only private insurance-company staff. C. Only foreign diplomatic missions stationed in India. D. Self-help groups (SHGs), NGOs and community-based organisations (CBOs).

Answer: D. Explanation:

- A: Incorrect: the source names community-based organisations, not only the armed forces, in this response layer.
- B: Incorrect: insurance staff are not named as search-and-rescue responders.
- C: Incorrect: diplomatic missions are not named as search-and-rescue responders.
- D: Correct: SHGs, NGOs and CBOs are named as involved in search and rescue alongside the community.

Trap: The named first-response actors are community, SHGs, NGOs and CBOs together, not the armed forces alone.

Q17. Which equipment is immediately useful in post-tsunami response?

A. Inflatable motorised boats, helicopters and search/rescue equipment. B. Long-range ballistic radar systems. C. Deep-sea drilling rigs. D. Agricultural harvesting combines.

Answer: A. Explanation:

- A: Correct: inflatable motorised boats, helicopters and search/rescue equipment are named as immediately required.
- B: Incorrect: ballistic radar systems are unrelated to the named immediate response equipment.
- C: Incorrect: drilling rigs are unrelated to tsunami response equipment.
- D: Incorrect: harvesting combines are unrelated to tsunami response equipment.

Trap: Immediate response equipment is specifically boats/helicopters/search-rescue gear, not unrelated heavy machinery.

Q18. What role did the Indian Naval Hydrographic Department perform during the 2004 response?

A. It transferred all its ships to a foreign navy for the response period. B. It deployed seven survey ships to open sea lines of communication and re-chart bathymetry. C. It deployed seven survey ships solely for tourism-promotion purposes. D. It grounded its entire survey fleet for the duration of the response.

Answer: B. Explanation:

- A: Incorrect: the ships remained under Indian naval deployment for the response, not transferred abroad.
- B: Correct: INHD deployed seven survey ships to reopen sea lines of communication and re-chart bathymetry after 2004.
- C: Incorrect: the deployment's purpose was operational response, not tourism promotion.
- D: Incorrect: the source describes active deployment, not grounding of the fleet.

Trap: The seven-ship deployment served operational sea-lane and bathymetry re-charting purposes, not an unrelated function.

Q19. From which networks does India's tsunami-warning seismic subsystem receive real-time data?

A. Only from foreign navies' classified sensor arrays. B. Only from state-level revenue department field offices. C. From IMD's national seismic network and international networks. D. Only from a single privately operated seismic station.

Answer: C. Explanation:

- A: Incorrect: classified foreign-navy sensors are not the named data source.
- B: Incorrect: revenue department offices are unrelated to seismic-data supply.
- C: Correct: the seismic subsystem draws real-time data from IMD's national seismic network and international networks.
- D: Incorrect: the source names a national-plus-international network, not one private station.

Trap: The seismic subsystem's data sources are IMD's national network plus international networks together, not one isolated source.

Q20. Which planning decisions can TUNAMI-N2 inundation scenarios inform?

A. Only income-tax assessment decisions in coastal districts. B. Only crop-insurance premium decisions for inland farmers. C. Only university-admission decisions for coastal-state students. D. Evacuation, settlement-avoidance and structural-design decisions.

Answer: D. Explanation:

- A: Incorrect: tax assessment is unrelated to the model's named use.
- B: Incorrect: crop-insurance premiums are unrelated to the model's named use.
- C: Incorrect: university admissions are unrelated to the model's named use.
- D: Correct: the TUNAMI-N2 model informs evacuation, settlement-avoidance and structural-design decisions.

Trap: The TUNAMI-N2 model's named use is risk-informed evacuation/settlement/design decisions, not unrelated administrative decisions.

Q21. Which ministry leads the model techno-legal framework for tsunami-safe zoning, planning, design and construction?

A. The Ministry of Home Affairs' (MHA) model techno-legal framework. B. The Ministry of Railways' track-safety framework. C. The Ministry of Civil Aviation's airspace framework. D. The Ministry of Textiles' industrial-safety framework.

Answer: A. Explanation:

- A: Correct: MHA's model techno-legal framework is named for tsunami-safe zoning, planning, design and construction.
- B: Incorrect: a railway track-safety framework is unrelated to tsunami-safe coastal construction.
- C: Incorrect: an airspace framework is unrelated to tsunami-safe coastal construction.
- D: Incorrect: an industrial-safety framework for textiles is unrelated to tsunami-safe coastal construction.

Trap: The named techno-legal framework is led by MHA, working alongside CRZ 2019 controls, not an unrelated ministry.

Q22. How many CRZ categories, from CRZ-I to CRZ-IV, structure coastal regulation?

A. A single, undivided coastal-regulation category applied uniformly. B. Four category-specific classes, CRZ-I to CRZ-IV, each with distinct controls. C. Two categories only, with no further sub-classification. D. Six categories, extending beyond CRZ-IV.

Answer: B. Explanation:

- A: Incorrect: the notification is explicitly category-specific, not a single undivided class.
- B: Correct: CRZ Notification, 2019 uses four category-specific classes, CRZ-I to CRZ-IV.
- C: Incorrect: the notification uses four categories, not two.
- D: Incorrect: the notification's named categories stop at CRZ-IV, not six.

Trap: Keep the exact category count (four: CRZ-I to CRZ-IV) distinct from an invented smaller or larger number.

Q23. Which two operational time figures for ITEWC must be kept distinct?

A. About 20 minutes for detection and about 10 minutes for dissemination. B. About 60 minutes for detection with no separate dissemination figure given. C. About 10 minutes for tsunamigenic-earthquake detection and about 20 minutes for advisory dissemination. D. About 10 minutes for both detection and dissemination, treated as one identical figure.

Answer: C. Explanation:

- A: Incorrect: this reverses the correct order of the two figures.
- B: Incorrect: the source names 10 and 20 minutes specifically, not a single 60-minute figure.
- C: Correct: detection is about 10 minutes and advisory dissemination is about 20 minutes -- two distinct stages.
- D: Incorrect: collapsing both stages into one identical figure erases the detection-to-dissemination gap.

Trap: Detection speed and advisory dissemination speed are two separate, sequential figures, not one interchangeable number.

Q24. Where is ITEWC housed, and which institute hosts it?

A. At the Geological Survey of India office, Kolkata. B. At the India Meteorological Department headquarters, New Delhi, as a standalone unit. C. At the Andaman and Nicobar administration office, Port Blair. D. At INCOIS, Hyderabad.

Answer: D. Explanation:

- A: Incorrect: the Geological Survey of India office in Kolkata is not ITEWC's location.
- B: Incorrect: IMD supplies seismic-network data but ITEWC itself is housed at INCOIS, not as a standalone IMD unit.
- C: Incorrect: the Port Blair administration office is not ITEWC's host location.
- D: Correct: ITEWC is housed at INCOIS, Hyderabad.

Trap: ITEWC's host institute and city (INCOIS, Hyderabad) must not be swapped for a different agency or city.

Q25. Which statement correctly frames UNESCO-IOC Tsunami Ready recognition?

A. It is a community-level, indicator-based recognition programme, not a national certification. B. It is a national certification automatically granted to every coastal country. C. It is a private insurance-industry rating system unrelated to UNESCO-IOC. D. It is a one-time award requiring no further maintenance of preparedness indicators.

Answer: A. Explanation:

- A: Correct: Tsunami Ready is a community-level, indicator-based UNESCO-IOC recognition, not a national certification.
- B: Incorrect: it is not automatically granted; it is community-specific and indicator-based.
- C: Incorrect: it is a UNESCO-IOC programme, not a private insurance rating.
- D: Incorrect: recognition requires maintained preparedness against defined indicators, not a one-time, no-upkeep award.

Trap: Tsunami Ready recognition is community-level and indicator-maintained, never a blanket national certification.

Q26. Which districts host India's first Tsunami Ready-recognised communities?

A. East Godavari and Krishna, both in Andhra Pradesh. B. Ganjam and Jagatsinghpur, both in Odisha. C. Kanyakumari and Nagapattinam, both in Tamil Nadu. D. Puri and Balasore, both in Odisha.

Answer: B. Explanation:

- A: Incorrect: these Andhra Pradesh districts are not the named locations of the first recognitions.
- B: Correct: Venkatraipur is in Ganjam district and Noliasahi is in Jagatsinghpur district, both in Odisha.
- C: Incorrect: these Tamil Nadu districts are not the named districts for India's first recognitions.
- D: Incorrect: these are different Odisha districts, not the two the source names.

Trap: Match the exact community-district-state triple; do not swap in a plausible-sounding but unnamed coastal district.

Q27. Why is an ordinary tall building not automatically a safe vertical-evacuation refuge?

A. Because vertical evacuation is prohibited for every coastal settlement. B. Because only underground shelters may be used after an earthquake. C. Because a refuge requires site-specific earthquake, hydrodynamic, debris, scour, capacity and accessibility design plus official designation. D. Because any building higher than two storeys is expected to collapse in every tsunami.

Answer: C. Explanation:

- A: Incorrect: vertical evacuation is appropriate where safe high ground is too distant.
- B: Incorrect: underground shelter is not the tsunami-evacuation rule.
- C: Correct: structural performance, capacity, access and official designation distinguish a refuge from an ordinary tall building.
- D: Incorrect: height alone neither proves failure nor proves safety.

Trap: Visible height is not a substitute for engineered and officially designated vertical-evacuation capacity.

Q28. Who should issue the local evacuation all-clear after a tsunami threat?

A. Any social-media account carrying a screenshot of an early bulletin. B. A Bottom Pressure Recorder automatically, without institutional review. C. ITEWC alone, regardless of local observations and authority. D. The competent local authority, using ITEWC updates and local conditions.

Answer: D. Explanation:

- A: Incorrect: unverified messages are not authoritative protective decisions.
- B: Incorrect: a BPR is a sensor, not a decision-making body.
- C: Incorrect: ITEWC supplies technical advisories but does not replace local protective authority.
- D: Correct: the competent local authority translates updated scientific information and local conditions into the all-clear.

Trap: A regional scientific advisory informs the decision; it is not itself the local all-clear.

Q29. How should the figure '13 tsunamis in 300 years, three in Andaman and Nicobar' be used?

A. As a document-period historical figure, requiring a current INCOIS/GSI source before being cited as an unchanging present-day statistic. B. As a permanently fixed statistic needing no further verification at any future date. C. As a figure describing only the twenty-first century's tsunami history. D. As a figure that excludes the Andaman and Nicobar region entirely.

Answer: A. Explanation:

- A: Correct: the source flags this as a document-period figure needing a current source before being treated as an unchanging present statistic.
- B: Incorrect: the source explicitly warns against treating it as permanently fixed with no further check.

- C: Incorrect: the figure covers a 300-year historical window, not only the twenty-first century.
- D: Incorrect: the figure explicitly includes 3 events in the Andaman and Nicobar region.

Trap: Historical counts recorded in a source document still need a current-source check before being quoted as today's figure.

Q30. Why does low-frequency, high-consequence tsunami risk require a different readiness strategy from recurrent hazards?

A. Because tsunamis are described as occurring more frequently than cyclones in India. B. Because rare events make sustained institutional readiness (monitoring, drills, retrofitting) harder to maintain politically and financially than for annually recurring hazards, even though consequence severity can exceed them. C. Because low-frequency hazards are defined as requiring no monitoring investment at all. D. Because cyclones and floods are described as having no consequence severity worth comparing.

Answer: B. Explanation:

- A: Incorrect: the analysis explicitly treats tsunamis as low-frequency compared with more frequently recurring hazards like cyclones.
- B: Correct: rarity makes sustained institutional readiness harder to maintain than for recurring hazards, despite potentially higher consequence severity.
- C: Incorrect: the analysis argues readiness still needs sustained investment, not that none is required.
- D: Incorrect: the comparison is about maintaining readiness, not denying cyclone/flood consequence severity.

Trap: The low-frequency/high-consequence argument is about the difficulty of maintaining readiness between rare events, not about denying any hazard's severity.

Q31. What institutional pattern often follows a catastrophic tsunami before readiness gradually erodes?

A. A complete transfer of all tsunami-warning functions to private commercial weather companies. B. An automatic doubling of the coastline's population within the same year as the event. C. A post-event institution-building spike (such as ITEWC's creation after 2004), followed by gradual readiness decay until the next event. D. An immediate and permanent end to any further institutional change once one warning centre is built.

Answer: C. Explanation:

- A: Incorrect: ITEWC is a public, multi-agency institution; the source does not describe a transfer to private commercial firms.
- B: Incorrect: population change is not the pattern the risk-system analysis describes.
- C: Correct: the Advanced analysis names a post-event institution-building spike followed by gradual readiness decay as the recurring pattern.
- D: Incorrect: the analysis argues institutional attention itself tends to erode over time, not remain permanently fixed after one build-out.

Trap: The readiness-decay pattern is about institutional attention rising after a disaster and fading before the next one, not a one-time permanent fix.

Q32. Which sea-level instruments must complement seismic detection when confirming a tsunami threat?

A. Only newspaper reports filed by local correspondents after the event. B. Only social-media trend analysis conducted after public reports of flooding. C. Only anecdotal fisherfolk radio chatter with no instrument backing. D. Bottom Pressure Recorders and tide gauges providing open-ocean and coastal sea-level observations.

Answer: D. Explanation:

- A: Incorrect: newspaper reports are not the named instrumental confirmation layer.
- B: Incorrect: social-media analysis is not the named instrumental confirmation layer.
- C: Incorrect: unverified radio chatter is not the named instrumental confirmation layer.

- D: Correct: Bottom Pressure Recorders and tide gauges supply the open-ocean and coastal sea-level observations that confirm wave generation alongside seismic analysis.

Trap: Confirmation of a tsunami bulletin rests on named instruments (BPRs and tide gauges) working with seismic analysis, not on unverified secondary reports.

PYQS AND ANSWER PRACTICE

Verified PYQ coverage

The 2025 GS-I tsunami-formation question is used for the disaster-management application of detection, warning and evacuation. The 2023 GS-I coastline question is addressed only for coastal hazard preparedness, and the 2024 GS-III resilience question is applied through a tsunami-specific lens. The 2025 seawater-intrusion question is adjacent coastal-water content, not a tsunami question. No verified 2026 Prelims or Mains question routes directly to this topic.

PYQ DEMAND CARD 1 - 2025 GS-I

Demand: Explain tsunami formation and consequences. (Explain · 10 marks · about 150 words)

Status: This card focuses on the disaster-management application; the physical-process explanation is a GS-I Geography demand.

Model answer (145 words): A tsunami is a series of long waves generated by sudden displacement of a large water volume, most commonly vertical seabed movement in an undersea earthquake, but also landslides, volcanic activity or meteorite impacts. Generation is not impact: an undersea earthquake does not automatically produce a destructive tsunami, since trigger distance, magnitude, bathymetry and coastal geometry govern whether and how waves affect a coast. In deep water the waves travel fast with low amplitude; entering shallow water, they shoal, slow and grow in height, producing run-up, inundation and a damaging return flow. Consequences include coastal flooding, structural damage, loss of life and livelihoods, and disruption to ports, fisheries and tourism. India's Tsunami Early Warning Centre (ITEWC) at INCOIS integrates seismic analysis with sea-level observation to confirm generation and update threat assessments, but the local evacuation all-clear remains a competent-authority decision, not a warning-centre output alone.

Examiner note: Named evidence (seabed displacement, shoaling/run-up, ITEWC's detection-to-advisory chain) does the evidentiary work; do not invent an unsourced wave-height, speed or casualty figure.

Answer plan: trigger set and definition -> generation-is-not-impact distinction (bathymetry/distance) -> deep-water-to-shoaling mechanism -> named consequences -> one warning-chain closing line.

Improvement pointer: name the 2004 event only as a one-clause mechanism illustration, not a casualty recital, to keep the answer analytical rather than descriptive.

PYQ DEMAND CARD 2 - 2023 GS-I

Demand: Comment on the resource potential of India's coastline and highlight hazard preparedness. (Comment and highlight · 15 marks · about 250 words)

Status: This card addresses only the tsunami and coastal-preparedness dimensions of the wider coastline demand.

Model answer (234 words): India's coastline exceeds 2,200 km across the East and West (a document-period figure), carrying genuine resource potential in fisheries, ports, shipping, tourism and offshore energy, alongside recurring hazard exposure: 13 tsunamis on historical record over the last 300 years, three in the Andaman and Nicobar region, besides cyclones, storm surge and coastal erosion. Realising this potential safely depends on a linked preparedness chain rather than any single measure. Detection and warning: ITEWC at INCOIS integrates seismic analysis with open-ocean Bottom Pressure Recorders and coastal tide gauges, issuing staged bulletins in roughly 10 to 20 minutes as an approved UNESCO-IOC Tsunami Service Provider under IOTWMS. Mapping and regulation: the TUNAMI-N2 model and Naval Hydrographic Department bathymetry feed inundation maps, while the CRZ Notification, 2019's category-specific CRZ-I to CRZ-IV controls regulate risk-sensitive coastal development rather than imposing one universal setback. Protection: structural measures (sea walls, breakwaters, submerged sand barriers and dykes) work alongside ecosystem-based buffers (mangroves, coastal forests, sand dunes with sea-weed planting and bio-shields), since the 2004 tsunami showed heavier damage precisely where dunes were mined. Response and

recovery: community first response, SHG/NGO/CBO-supported search and rescue, and the Naval Hydrographic Department's seven-ship 2004 deployment illustrate multi-agency capacity. Last-mile safety closes the chain: accessible evacuation routes, drills and UNESCO-IOC 'Tsunami Ready' community recognition, beginning with Venkatraipur and Noliasahi in Odisha in 2020, convert coastline potential into genuinely resilient use rather than unmanaged exposure.

Examiner note: Keep resource potential (fisheries/ports/tourism) and hazard preparedness (ITEWC, CRZ, structural/ecosystem buffers, Tsunami Ready) as two linked but distinct halves of the answer.

Answer plan: open on resource potential -> pivot explicitly to hazard exposure -> sequence the preparedness chain (detection -> mapping -> protection -> response -> last-mile) -> close by linking potential to preparedness.

Improvement pointer: name at least one ecosystem-based measure (e.g. mangrove plantation) alongside a structural one, since balanced-mitigation knowledge is specifically rewarded here.

PYQ DEMAND CARD 3 - 2024 GS-III

Demand: Describe the elements that determine disaster resilience.

Directive and limit: Describe · 15 marks · about 250 words.

Status: This card applies the general disaster-resilience framework specifically to tsunami warning, land use, buffers, evacuation and community readiness.

Model answer (212 words): Disaster resilience is the capacity of a system or community to anticipate, absorb, respond to and recover from a hazard while preserving essential functions. In a tsunami context, six linked elements determine this capacity. First, scientific capability: ITEWC at INCOIS must combine seismic analysis, models, Bottom Pressure Recorders and tide gauges to issue staged advisories. Second, risk knowledge: TUNAMI-N2 scenarios, bathymetry and local topography should guide inundation maps, evacuation zones and critical-facility siting. Third, risk-sensitive development: CRZ 2019 categories and local land-use controls must limit new exposure. Fourth, layered protection: engineered shelters, retrofits or breakwaters should be combined with dunes, mangroves, reefs and other ecosystem buffers; the 2004 damage pattern where dunes were mined shows why degradation weakens resilience. Fifth, inclusive last-mile capacity: multilingual warnings, accessible horizontal or designated vertical evacuation, repeated drills and support for fishers, tourists, older persons and persons with disabilities convert an advisory into safe action. Sixth, institutional continuity: UNESCO-IOC Tsunami Ready recognition, first achieved in India by Venkatraipur and Noliasahi in 2020, must be renewed so rare-hazard readiness does not decay. Recovery should then restore ports, fisheries, housing and services through Build Back Better. Thus resilience is produced by a maintained chain from knowledge and prevention to local action and adaptive recovery, not by one warning technology.

Answer plan: define resilience -> organise the six tsunami-specific elements -> attach 2004 and Tsunami Ready evidence -> conclude with the maintained-chain test.

Improvement pointer: keep the answer disaster-specific while showing that scientific, social, institutional, ecological and recovery capacities operate together.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish tsunami generation from tides and cyclone-driven storm surge. Answer in about 150 words.

Model answer (138 words): A tsunami is not an astronomical tide and not a cyclone-driven storm surge. Tides follow predictable gravitational cycles of the moon and sun; a storm surge is chiefly wind-and-pressure driven, building up during a cyclone's approach. A tsunami's generation instead begins with sudden displacement of a large water volume, most commonly vertical seabed movement during an undersea earthquake, though landslides, volcanic activity and meteorite impacts are alternative triggers. This distinction matters operationally: tide tables and cyclone-track forecasts cannot substitute for seismic and sea-level monitoring, since an undersea earthquake does not automatically produce a destructive tsunami -- source mechanism, depth, displacement, distance, bathymetry and coastal geometry determine whether and how waves actually affect a coast. Conflating the three phenomena would misdirect both warning systems and public understanding, since each requires a distinct detection method, lead time and institutional response chain.

Answer plan: negative definition first (not a tide, not a storm surge) -> driver-based distinction in prose -> tie back to operational consequence (wrong monitoring tool used).

Improvement pointer: name the specific instrument each phenomenon needs (tide tables; cyclone-track forecasts; seismic/sea-level monitoring) in one explicit sentence for a sharper close.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain the detection, forecast, warning and local all-clear chain for tsunamis. Answer in about 150 words.

Model answer (144 words): Tsunami detection, forecast, warning and the local all-clear are four distinct stages, not one collapsed step. Detection begins with seismic analysis identifying an undersea earthquake's parameters, cross-checked by open-ocean Bottom Pressure Recorders and coastal tide gauges confirming actual wave generation. India's Tsunami Early Warning Centre (ITEWC) at INCOIS, Hyderabad, operational since 15 October 2007, performs this detection in roughly 10 minutes. Forecast follows, using the TUNAMI-N2 model (customised by ICMAM and INCOIS) to estimate likely surge and inundation for the confirmed scenario. ITEWC then issues a staged bulletin -- threat category and advisory -- within roughly 20 minutes, functioning as an approved UNESCO-IOC Tsunami Service Provider under the IOTWMS for India and the region. The local all-clear, however, is not an ITEWC output: competent local authorities must weigh local conditions before declaring safety, since a regional advisory is evidence for a decision, not the decision itself.

Answer plan: state the four-stage claim upfront -> walk each stage in strict order -> close on the local-authority boundary.

Improvement pointer: cite the two time figures (roughly 10 minutes / 20 minutes) explicitly and distinctly, since examiners specifically reward not collapsing them into one number.

ORIGINAL MAINS 3 - 15 MARKS

Question: Discuss how India combines structural and ecosystem-based measures for coastal-hazard mitigation. Answer in about 250 words.

Model answer (233 words): Coastal-hazard mitigation combines structural and ecosystem-based measures, since the 2004 Indian Ocean tsunami showed neither works alone. Structural measures include cyclone shelters, submerged sand barriers and dykes, location-specific sea walls and breakwaters, plus identification of vulnerable structures for retrofitting; each targets a specific stretch of coast, not one universal design. Ecosystem-based measures include sand dunes planted with sea-weed, mangrove and coastal-forest plantation, bio-shields, and coral reefs -- natural buffers, some doubling as disaster-management sanctuaries with public-awareness spaces. The 2004 event showed mangroves, forests, dunes and cliffs provided the best natural barriers, while heavy damage occurred specifically where dunes had been mined, proving that removing an ecosystem buffer itself raises risk, not a neutral, cost-free choice. The CRZ Notification, 2019's category-specific CRZ-I to CRZ-IV controls, combined with tsunami inundation/hazard mapping and MHA's model techno-legal framework for zoning, planning, design and construction, give this combination a regulatory backbone. Treating either category as sufficient alone -- engineered walls without buffers, or buffers without retrofitted structures -- would leave a coast only partially protected against a low-frequency, high-consequence hazard. The M.S. Swaminathan Research Foundation, Pondicherry, functions as a local-knowledge centre linking community awareness to engineered design, while NDMA's vulnerability mapping keeps the structural side updated. A balanced answer names both categories together, cites the 2004 dune-mining evidence for why ecosystem removal is not cost-free, and roots the CRZ-plus-MHA backbone as the mechanism binding both into one coherent system.

Answer plan: open with the complementary-categories thesis -> list structural measures -> list ecosystem-based measures (including coral reefs) -> cite 2004 dune-mining evidence -> close on the regulatory backbone (CRZ+MHA).

Improvement pointer: state explicitly that ecosystem-based measures are not secondary to structural ones -- examiners reward this named distinction over a flat list of measures.

ORIGINAL MAINS 4 - 15 MARKS

Question: Analyse the institutional architecture and limits of India's tsunami detection-to-advisory system. Answer in about 250 words.

Model answer (221 words): India's tsunami detection-to-advisory architecture is genuinely capable but structurally multi-agency and resource-dependent. The Indian Tsunami Early Warning System was established with the Department of Space, the Department of Science and Technology and CSIR; its seismic subsystem draws real-time data from IMD's national seismic network, together with international networks, while Bottom Pressure Recorders and tide gauges confirm open-ocean wave propagation. ITEWC at INCOIS, Hyderabad, operational since 15 October 2007, integrates this evidence to issue bulletins in roughly 10 minutes for detection and 20 minutes for advisory dissemination, as an approved UNESCO-IOC Tsunami Service Provider under IOTWMS for India and the region. The Indian Naval Hydrographic Department regularly supplies bathymetry for inundation mapping, and famously deployed seven survey ships after 2004 to reopen sea lines of communication and re-chart the seabed, while the TUNAMI-N2 model, customised by ICMAM and INCOIS, converts this evidence into evacuation, settlement-avoidance and structural-design decisions. This capability's real limits are institutional rather than purely technical: sustaining seismic-station upkeep, periodic bathymetric resurveying and drilled community response between rare events is harder to maintain politically and financially than for annually recurring hazards such as cyclones, so continued institutional discipline -- not one-time technology deployment -- determines whether fast detection actually converts into last-mile local safety, and the local all-clear decision must still rest with competent local authorities rather than the warning centre itself.

Answer plan: name the multi-agency origin -> trace the seismic-to-advisory chain with the two time figures -> name the mapping/model layer -> close on institutional (not technical) limits.

Improvement pointer: end with an explicit sentence assigning the all-clear decision to local authorities, not ITEWC, to close the institutional-boundary argument being tested.

ORIGINAL MAINS 5 - 20 MARKS

Question: Critically examine UNESCO-IOC 'Tsunami Ready' recognition as an indicator of India's coastal resilience. Answer in about 250 words.

Model answer (247 words): UNESCO-IOC's 'Tsunami Ready' recognition is a community-level, indicator-based preparedness programme, not a national certification, and this distinction should anchor any evaluation of India's coastal resilience. Venkatraipur (Ganjam) and Noliasahi (Jagatsinghpur) in Odisha became India's first recognised communities on 7 August 2020, having demonstrated maintained warning-relay, mapped evacuation-route and drill indicators specific to their locality, not a statewide or national outcome. This granular design suits a low-frequency, high-consequence hazard: a national-level claim would obscure the last-mile reality that safety depends on trusted local relay, accessible evacuation infrastructure, inclusive assistance for fisherfolk, harbour workers and tourists, and repeated drills -- none guaranteed by ITEWC's 10-to-20-minute detection chain alone. The risk is that recognising two communities is quietly generalised into an unverified claim of coast-wide readiness, since institutional attention to rare hazards spikes after a disaster and decays before the next one. A sound evaluation should instead track how many more communities are recognised over time, whether recognition is periodically renewed rather than a one-time badge, whether indicators match observed drill performance rather than the ITEWC bulletin chain alone, and whether structural retrofitting, mangrove/dune protection and CRZ-linked land-use discipline advance alongside the recognition programme. The benchmark indicators -- hazard mapping, warning dissemination, public education, periodic drills -- work only where local governance sustains them between disasters; a 2020 badge says nothing about whether relay equipment still functions or new arrivals have been briefed since. Genuine coastal resilience is best read as a renewable, continuously maintained process, not a fixed, one-time national status.

Answer plan: state the community-vs-national boundary as the thesis -> give the named evidence (Venkatraipur/Noliasahi, 2020) -> argue why granularity suits a low-frequency hazard -> name the generalisation risk -> close with renewal-based criteria for genuine resilience.

Improvement pointer: add one explicit recommendation (e.g. publish an annual renewed-recognition count) to convert the critique into an actionable improvement, which 20-mark 'critically examine' answers are specifically expected to supply.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design an exam-safe tsunami-resilience framework covering detection, land use, protection, evacuation and recovery. Answer in about 250 words.

Model answer (245 words): An exam-safe tsunami-resilience framework separates five linked layers, not one blurred claim. First, detection: ITEWC/INCOIS's seismic-plus-sea-level chain (Department of Space, DST, CSIR; Bottom Pressure Recorders) delivers roughly 10-minute detection and 20-minute advisory dissemination as an approved UNESCO-IOC Tsunami Service Provider. Second, mapping and land use: TUNAMI-N2 model outputs and Naval Hydrographic Department bathymetry feed inundation maps, matched to the CRZ Notification, 2019's category-specific CRZ-I to CRZ-IV controls and MHA's techno-legal framework. Third, mixed protection: structural measures (sea walls, breakwaters, sand barriers/dykes, retrofitted structures) alongside ecosystem-based buffers (mangroves, coastal forests, dunes, bio-shields, coral reefs), since 2004 showed heavier damage where dunes were mined. Fourth, evacuation readiness: accessible warning relay, mapped routes, community first response with SHG/NGO/CBO support, drills and UNESCO-IOC 'Tsunami Ready' recognition, renewed rather than permanent. Fifth, recovery: restoring ports, fisheries and tourism via Build Back Better siting, supported by inflatable boats, helicopters and search-rescue equipment immediately post-event. Local all-clear authority stays with competent local administration throughout, since no single layer alone proves last-mile safety. A sixth, governance thread cuts across all five: State and District Disaster Management Authorities under the Disaster Management Act, 2005 must translate NDMA's guidelines into locally funded, evaluated and accountable action, since a rare hazard otherwise suffers readiness-decay as budgets and drills quietly lapse. A full-marks answer sequences these six threads, cites 2004 as ITEWC's institutional origin point, and closes by naming continuous inter-agency coordination -- not one technology or badge -- as the true measure of resilience.

Answer plan: name the five layers upfront as an explicit list -> develop each layer in one tight sentence with named evidence -> add the cross-cutting governance thread -> close by naming the single true measure of resilience.

Improvement pointer: state which single layer would fail first without institutional discipline (readiness-decay) to show evaluative depth beyond a descriptive five-layer list.